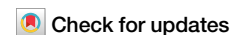




Geographic and perceptual bias in multimodal LLMs: evidence from a global dataset spanning more than 200 cities



Hyebin Kim¹, Xuan Li¹, Matias Quintana², Filip Biljecki^{3,4} & Sugie Lee¹ ✉

Multimodal large language models (MLLMs) are increasingly deployed for urban informatics applications, from objective built environment attribute extraction to subjective assessments. However, do MLLMs apply the same standards across regions while executing these urban-related tasks? Are MLLMs reliable and neutral judges? Do they adapt without criticizing, raising concerns about equitable deployment and further hidden bias? Here, we present a geographic red-teaming framework for diagnosing geographic biases in MLLMs applied to image object detection and urban visual perception. Using annotated imagery from more than 200 cities as a standardized benchmark, we demonstrate that MLLMs exhibit significant geographic disparities in both object detection accuracy and perceptual assessments across different regions. Quantitatively, perceptual scores for attributes such as Wealthy and Beautiful dropped by an average of 80% for African cities after geo-referencing, whereas regions such as Asia and North America showed score increases of 26%. A spatial-scale sensitivity test further showed that coordinates, city, country, and continent information produced broadly similar directional shifts, while combined geographic cues generated the strongest perceptual changes. Among the multiple state-of-the-art models, GPT-4o showed the lowest perceptual bias. Our geographic red-teaming stress-tests MLLM performance across underrepresented urban environments, revealing consistent patterns of bias that mirror training-data imbalances. We also provide quantitative metrics for measuring geographic equity in model outputs and establish recommendations for responsible MLLM deployment in urban research. These findings highlight critical limitations in current multimodal AI systems and demonstrate the urgent need for geographically inclusive model development to prevent the perpetuation of urban inequalities through automated analysis systems.

Large Language Models (LLMs), particularly in their multimodal form, are increasingly being adopted in urban informatics¹. Their text-based reasoning capabilities have proven effective for tasks such as personalized travel planning², disaster impact assessment³, and urban planning support⁴. Building on these advances, recent Vision-Language Models now enable complex visual analytics in urban contexts. Application of Multimodal Large Language Models (MLLMs) were rapidly developed from extracting, describing, reasoning about, or perceiving urban environments^{5,6}. Earlier work using LLMs used street images as a widely existed data source in global cities to predict societal factors like population density, accessibility to

healthcare⁷, and built environment auditing⁸. For instance, OpenFACADES leverages MLLMs to enrich building profiles with both objective attributes and semantic descriptors, such as building type, surface material, number of floors, and building age⁹. Applications on more subjective metrics include analyzing walkability¹⁰, attractiveness¹¹, visual quality¹², and safety¹³.

Overall, MLLMs have many advantages such as their ease of use and generability. Compared to usually conventional CNN-based models, MLLMs can be scaled to predict new targets with a simple few-shots prompt. With explicit reasoning through generated text, MLLMs are sometimes regarded as more interpretable than traditional black-box models like

¹Department of Urban Planning and Engineering, Hanyang University, Seoul, Korea. ²Future Cities Lab Global, Singapore-ETH Centre, Singapore, Singapore.

³Department of Architecture, National University of Singapore, Singapore, Singapore. ⁴Department of Real Estate, National University of Singapore, Singapore, Singapore. ✉e-mail: sugielee@hanyang.ac.kr

CNN¹⁴, even though it is still controversial if the generated text is how LLMs actually think and reason to reach the final answer¹⁵.

Some recent studies have explored the substitution of human participants with LLMs. They are convincing particularly for labor-intensive tasks such as labeling building attributes⁹, however, the application of LLMs to more subjective concepts, such as perceived safety or aesthetic quality, raises both technical and ethical concerns¹¹. Even though comparing the LLM-generated results against expert or crowd perception is a necessity for these types of work to be accepted through peer review, there remains a lack of consensus regarding appropriate evaluation protocols, especially the criteria by which LLM-generated responses should be accepted or rejected in practice.

Are LLMs neutral judges? To what extent LLMs produce biased results? How does the potential geographic bias vary in the context of urban studies? These questions underpin a growing body of research focused on the sociotechnical implications of LLMs and highlight the urgent need to interrogate how these models may perpetuate, or even exacerbate, existing inequalities¹⁶. Several studies have already documented the presence of social biases within the field of Natural Language Processing (NLP)^{17–19}. These biases encompass a wide range of socio-demographic dimensions, including gender bias^{20,21}, racial bias^{22–24}, age bias^{25,26}, and ethnic bias^{27,28}, as well as geographic bias^{29,30} and cultural bias^{31,32}. As commonly noted across recent studies, biases in LLMs originate primarily from their training datasets.

The geography bias and limited transferability to data-scarce or underrepresented regions from textual content generated by LLMs have been proved^{29,33,34}. Lower error rates have been reported for Western and higher-income countries through testing against the World Bank benchmark³⁴. In some case studies of United States, areas characterized by lower population density, higher percentages of white residents, and lower income levels were less likely to receive location-specific responses from ChatGPT³⁵; Jang and Kim⁸ also noticed geographic disparities in the accuracy of building environment auditing from Gemini.

Few studies have investigated whether explicit geographic metadata such as coordinates, city names, or regional labels influence LLM behavior^{30,33,36}. Yet these works remain limited in generalizability, and such an oversight constrains our understanding of how contextual cues influence model outputs, particularly in perception tasks requiring spatial sensitivity. Moreover, there remains a notable absence of standardized benchmarks or prompt-engineering techniques specifically designed to test geographic generalizability in diverse urban contexts.

In the field of LLM post-training, Red-teaming for LLMs is a practice originally rooted in cybersecurity and military strategy, aiming to the stress-testing of systems to systematically uncover vulnerabilities and expose unsafe behaviors^{37–39}. Existing red-teaming efforts have established comprehensive review-and-revision mechanisms to address human-centric discrimination across dimensions¹⁶, lacking in domain-specific tasks like urban perception, regrade less the popular use of LLM in research and practice^{10,40}. More attentive and critical application of LLM-based methods is essential to reveal disproportionate representation and misuse in practice, particularly for downstream applications where LLMs plays an important role of auditing or judge⁸.

This study poses three central research questions. First, do MLLMs show differences in image instance detection across different regions? Second, does geographic information alter the MLLMs' visual perceptions on urban environments across regions? Lastly, what advice we can provide to model using and model training through the Geographic red-teaming?

Our study addresses the LLM bias in urban perception by proposing a Geographic Red-teaming framework. This geographic red-teaming workflow systematically assesses spatial bias in MLLMs by utilizing a global dataset of annotated street-view imagery from 204 cities across six regions⁴¹. Using zero-shot prompting on three MLLMs (GPT-4o Gemini 2.0 Flash, and Claude 3.7 Sonnet) the framework compares model performance under two experimental conditions: non-referenced (visual input only) and geo-referenced (visual input supplemented with city and country labels). The

models are evaluated on two primary tasks: physical object detection for elements like cars and buildings, and urban perception ratings for subjective qualities such as safety and wealth. By analyzing disparities in detection accuracy and shifts in perceptual scores across different global regions, we quantify geographic bias, providing a diagnostic report on the model's fairness and spatial sensitivity in urban informatics (Fig. 1). By doing so, we supplement existing red-teaming strategies, provide a diagnostic approach to identify geographic bias in multimodal AI systems, and give advice for users to mitigate the bias in operation.

As a result, we focus on two key analytical tasks in urban planning field: instance detection and perceptual attribute assessment, each targeting different dimensions of LLM performance. Building on representativeness and prior studies, three MLLMs are evaluated in this study: GPT-4o from OpenAI, Gemini 2.0 Flash from Google, and Claude 3.7 Sonnet from Anthropic.

Results

Overview of model performance

There is two major ways of using MLLM on urban tasks: instance detection and perceptual attribute assessment, each targeting different dimensions of LLM performance. As for the instance detection, which evaluates the models' ability to identify ten objective physical features (e.g., cars, buildings, vegetation), the primary metric is the agreement score, which quantifies the alignment between the objects detected by the MLLMs and the validated reference labels from the Global Streetscapes benchmark. As for perceptual tasks, geographic bias is assessed by comparing model outputs under two conditions: non-geo-referenced (visual input only) and geo-referenced (visual input supplemented with location metadata). By observing the shift in scores between these conditions, we can isolate and quantify the influence of geographic identifiers on model judgment. A comprehensive description of the dataset, prompt design, and mathematical derivation of these metrics is provided in the Methodology section.

An initial comparison of runtime efficiency, output formatting accuracy, and operational cost revealed clear distinctions among the three evaluated MLLMs. We selected a representative state-of-the-art model each from each of the leading MLLM providers: GPT-4o from OpenAI, Gemini 2.0 Flash from Google, and Claude 3.7 Sonnet from Anthropic. Gemini 2.0 emerged as the fastest and most cost-efficient model across both tasks. However, this performance advantage came with trade-offs. Gemini also produced the highest number of malformed or invalid responses, particularly in the instance detection task, with 369 error cases out of 6000 images, compared to zero from Claude and OpenAI (Table 1). Malformed or invalid responses refer to cases where the model either failed to understand the given image and thus provided an invalid output, or produced a response containing explanatory text instead of the required numbers and commas as specified in the prompt.

In terms of processing time, both Claude and OpenAI demonstrated comparable durations, averaging around 200% of the baseline for each task, which involved instance detection and perceptual attribute evaluation across 6000 images. However, Claude incurred the highest total cost as it approximately costed over 2000% of the baseline across both tasks, while OpenAI remained moderately priced, and Gemini was by far the least expensive. These findings highlight the need to balance computational efficiency and economic feasibility against model robustness and formatting fidelity.

2. Instance detection

Each MLLM was prompted with the same image and instructed to detect instances of urban features. Agreement scores represent mean accuracy across all 6000 images per model. We visualized the overall agreement scores by country for each model and the delta between each model's country-level score (Fig. 2a). Summarizing the results across all continents, the top five countries with the highest scores from Claude were Brunei (1.00, Asia), British Virgin Islands (1.00, North America), and Iraq (0.95, Asia). In contrast, the five countries with the lowest scores were Nicaragua (0.70,

Data Preprocessing

NUS Global Streetscapes

Temporal	Hour
Source	Season
Weather	Platform
Panoramic	



Model Analysis

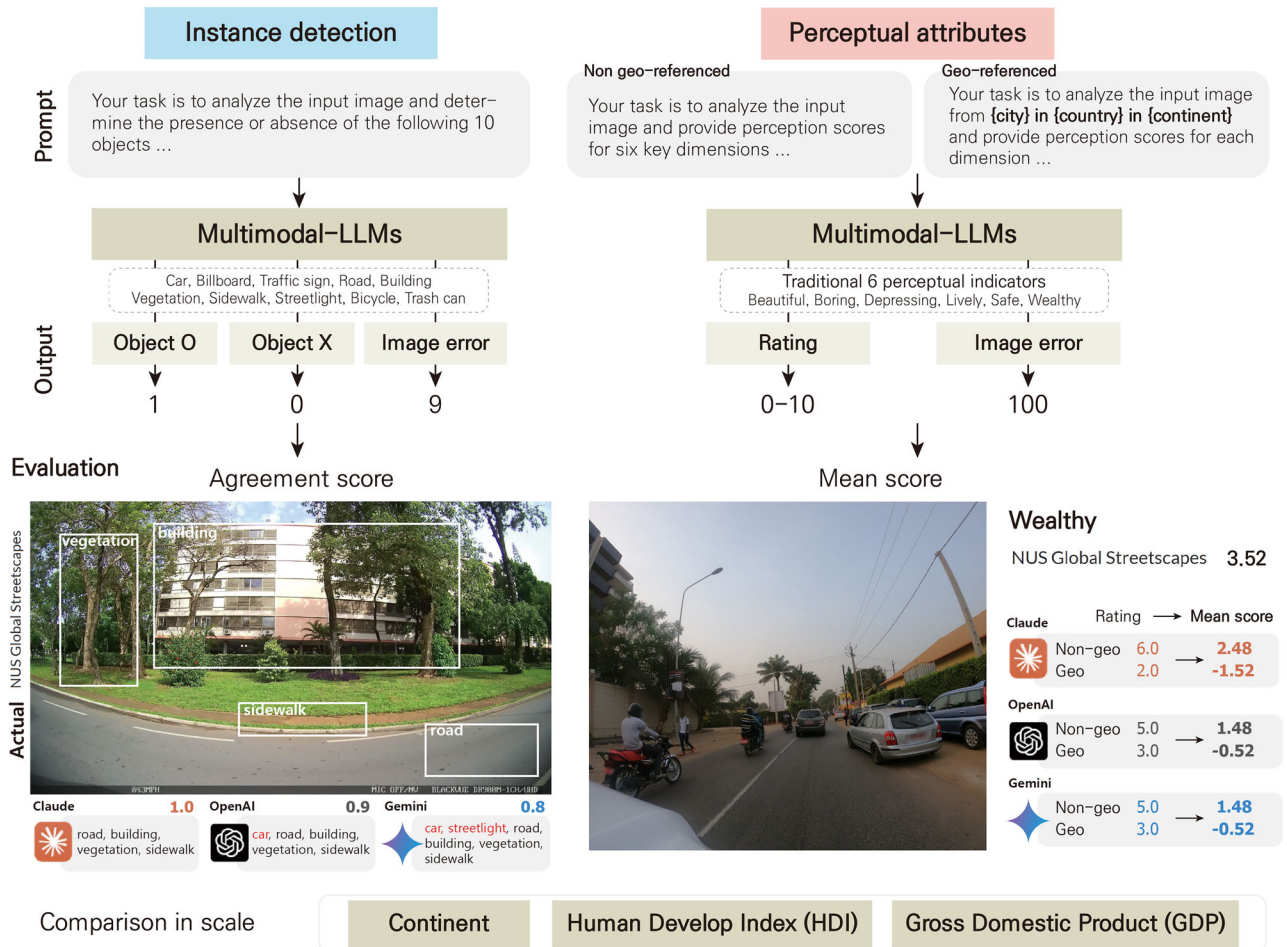


Fig. 1 | Workflow for MLLM analysis of Global Streetscapes. Street-view image examples were obtained from the Global Streetscapes dataset.

Table 1 | Overview of performance

Criterion	Instance detection			Perceptual attributes					
				Non Geo-referenced			Geo-referenced		
	Claude	OpenAI	Gemini	Claude	OpenAI	Gemini (baseline)	Claude	OpenAI	Gemini
# of Error response	0	0	369	0	2	19	0	2	0
Processing time (s)	206	247	138	215	179	100	183	190	109
Cost (USD)	2640	1510	95	2650	1620	100	2350	1490	99

*Note: # of Error responses indicates the number of failed responses (mis-formatted or invalid) among 6000 times of request. Processing time and cost are normalized relative to the Gemini model (Gemini = 100).

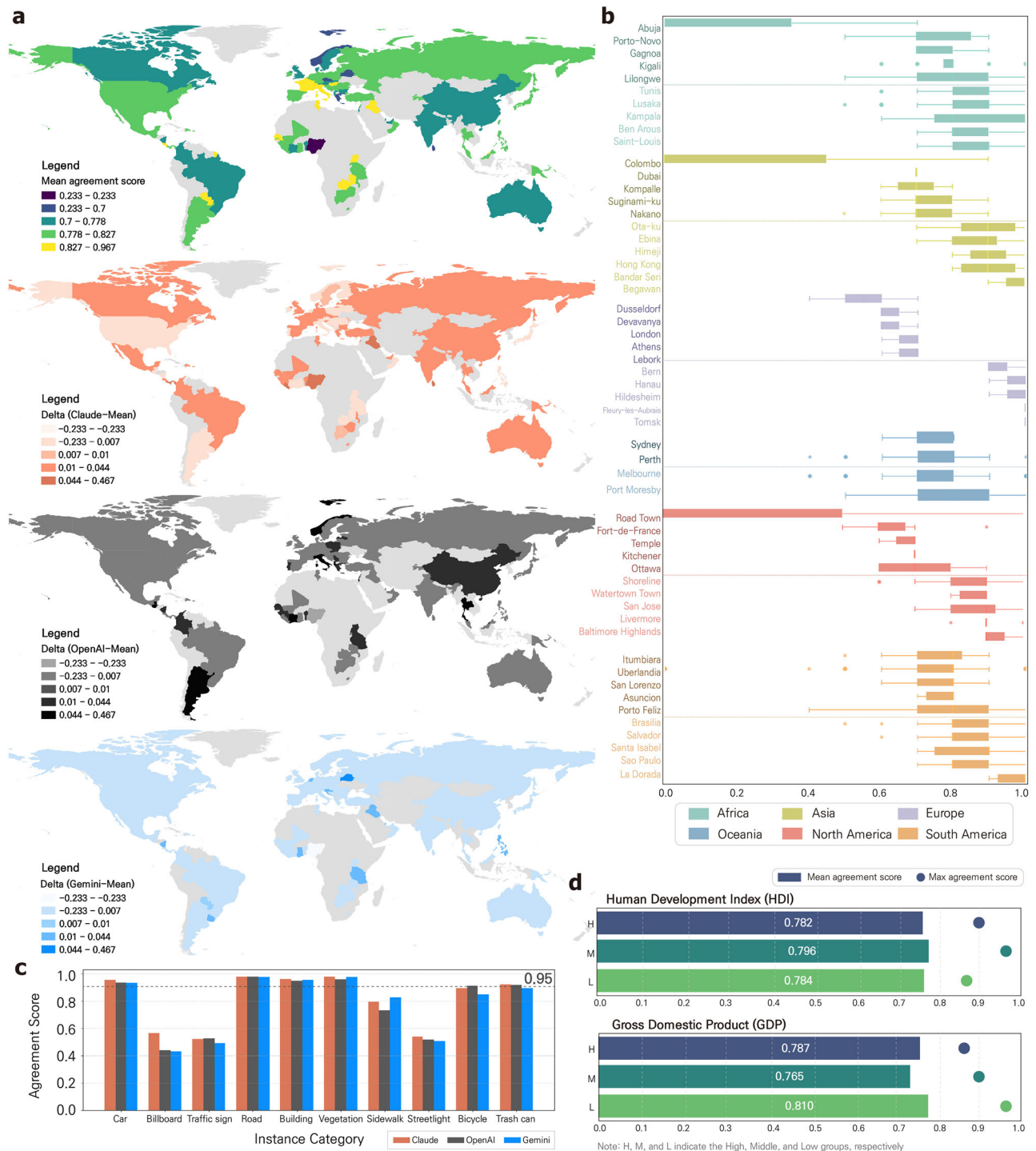


Fig. 2 | Instance detection performance of MLLMs across geography and object types. **a** Global maps show country-level mean agreement scores and model-specific deviations from the cross-model mean. Positive delta values indicate above-average performance, while negative values indicate below-average performance. **b** City-level agreement scores are summarized by continent. Colors indicate continents. **c** Object-level agreement scores are compared across the ten instance

categories for Claude, OpenAI, and Gemini. **d** Mean agreement scores are compared across Human Development Index and Gross Domestic Product groups. H, M, and L indicate High, Middle, and Low groups, respectively, based on tertile classification within the analytical sample. MLLM denotes multimodal large language model; HDI denotes Human Development Index; GDP denotes Gross Domestic Product.

North America), Albania (0.70, Europe), and Norway (0.60, Europe), indicating generally higher performance in Asia and North America. For OpenAI's GPT-4o, the highest-performing countries and territories were Brunei (1.00, Asia), Hong Kong (0.95, Asia), West Bank (0.90, Asia). On the other hand, the lowest scores were found in Sri Lanka (0.45, Asia), Nigeria (0.00, Africa), and the British Virgin Islands (0.00, North America). Lastly,

Gemini showed strong performance in Brunei (0.90, Asia), Iraq (0.90, Asia), and Hong Kong (0.90, Asia). In contrast, it recorded the lowest scores in Sri Lanka (0.45, Asia), the British Virgin Islands (0.00, North America), and Nigeria (0.00, Africa). In terms of the agreement scores across models, Claude achieved the highest overall agreement score (0.810), followed by OpenAI (0.786) and Gemini (0.783), indicating superior detection

consistency across urban features. Regarding the delta values, higher values indicate a positive bias, while lower values indicate a negative bias. The results show that although the overall bias levels were similar across models, no countries exhibited markedly large bias differences among the three models.

In addition, detailed agreement scores by city were visualized in Fig. 2b using bar plots, highlighting the top and bottom five cities in each continent. Interestingly, higher average agreement scores were observed in Africa and North America, with Claude achieving 0.835 in Africa and 0.832 in North America. These results contradict prior assumptions that models may perform worse in underrepresented or marginalized regions. This observation partially challenges existing literature, which suggests that the Global South is more vulnerable to model error or hallucination^{29,33}. While this trend was not strongly reflected here, the result warrants further contextual interpretation, particularly when considering scene characteristics such as image clarity and composition.

To assess whether these continent-level patterns were broadly shared across cities, we additionally examined the distribution of city-level mean agreement scores within each continent (Figure S1 and Table S1 in the Supplementary Information). The analysis shows relatively narrow within-continent spreads overall, particularly for Claude, indicating that the continent-level instance detection patterns are not driven solely by a few outlying cities. OpenAI and Gemini show somewhat larger inter-city dispersion in several continents, but the overall continental differences remain modest.

Furthermore, model performance varied by object type (Fig. 2c). To obtain object-level results, agreement scores were aggregated across all images containing each object category, and the mean agreement score was computed for each model and category. Across all models, vegetation, road, and building were the most accurately detected features, consistently achieving mean agreement scores exceeding 0.95. Claude showed particularly strong performance in detecting trash cans (0.922) and cars (0.954). In contrast, all models struggled with billboards and traffic signs, where agreement scores fell below 0.57—likely due to image resolution limitations or visual ambiguity in these smaller features (Figure S2 in the Supplementary Information). These results suggest that while all models demonstrate general competence in object-level urban interpretation, Claude consistently provides more accurate outputs in objective tasks, reinforcing its potential value for structured urban built environment audit.

Lastly, we compared the mean agreement scores across groups categorized by the Human Development Index (HDI) and Gross Domestic Product (GDP) to examine whether model performance varies with broader socio-economic and developmental contexts (Fig. 2d). The results, similar to the continent and model-level comparisons, showed minimal differences across HDI groups. Regions with a medium HDI recorded a slightly higher mean score of 0.796, but the variation among groups was negligible. This suggests that a country's HDI does not appear to have a significant correlation with object detection performance. GDP also showed no striking differences overall. However, regions classified as having low GDP exhibited a relatively higher mean score of 0.810. This finding contrasts with prior studies that measured LLM performance against conventional socio-economic indicators, in which higher socio-economic status often correlated with better outcomes³⁴. In this case, object detection results demonstrated little to no relationship with HDI or GDP, with the exception that lower-GDP regions performed better. We propose that this pattern may be linked to factors such as the relative complexity of urban landscapes.

In addition, we conducted equivalence tests using the TOST procedure to assess whether the differences between HDI- and GDP-based groupings were statistically negligible (Table S2 in the Supplementary Information). The results provide statistical support that differences between HDI- and GDP-based classifications are generally negligible. At the group level, the Middle group consistently satisfies the equivalence condition, indicating stable agreement scores across grouping schemes. The High and Low groups show some mixed results, where not all cases meet the equivalence threshold; however, the magnitude of differences remains small overall.

These findings suggest that agreement scores are largely stable regardless of whether countries are grouped by HDI or GDP.

Perceptual attributes

Overall, we observed variation in perceptual scores before and after geographic information such as city, country, and continent was provided. On average, perceptual ratings in several regions, particularly Africa and South America, shifted in a more negative direction after geo-referencing (Fig. 3a, b). For example, in the Wealthy attribute, African countries often received neutral or slightly positive scores in the non-geo condition (Fig. 3a). However, when the geographic context was included in the prompt, these ratings dropped significantly (Fig. 3b). This shift suggests the presence of a negative perceptual bias linked to geographic identifiers.

Conversely, regions such as Asia, North America, and Oceania experienced positive shifts in perception when geo-information was introduced. Among all attributes, Safe showed the most pronounced shift between non-geo and geo conditions: while scores remained relatively modest across most regions without geographic context, the introduction of geo-referencing led to notably higher scores in Oceania and North America, and conversely more negative scores in parts of Africa and South America. The Lively attribute also exhibited a marked contrast between conditions, with East Asian countries—particularly China—shifting from strongly negative scores in the non-geo condition toward relatively higher scores once geographic context was provided. In contrast, attributes such as Boring, Depressing, and Wealthy showed smaller absolute differences between geo and non-geo conditions across most regions, suggesting that MLLMs hold weaker or less spatially consistent geographic biases for these perceptual dimensions. This pattern demonstrates a tendency among MLLMs to associate certain regions with more favorable or safer perceptions when geo-information is provided, especially those in Asia, Europe, North America. This result suggests that perceptual bias may be more semantically or culturally embedded than simply a function of visual information, such as with more favorable environmental aesthetics and socio-economic conditions⁴².

When examining the polarity of this bias, we found that positive attributes such as Beautiful, Lively, Safe, and Wealthy tended to receive higher scores when geo-information was included, particularly in Asia, North America, and Oceania—consistent with the patterns observed in Fig. 3a, b. Conversely, negative attributes like Boring and Depressing decreased in these regions, reinforcing the presence of a positive bias. On the other hand, Africa and South America generally exhibited the opposite pattern: lower scores for positive attributes and higher scores for negative ones when geo-referenced prompts were applied, indicating the presence of negative perceptual bias in these regions.

To further clarify the magnitude of these shifts, we examined continent-level mean scores across the six perceptual attributes (Fig. 3c). The results show that the effect of geo-referencing is not uniform across attributes, but several shifts are large enough to be meaningful. In particular, the Lively attribute shows one of the most consistent aggregate-level changes across continents: geo-referenced scores become less negative than non-geo-referenced scores in all regions, with especially large differences in Oceania (−1.25 vs. −2.56), South America (−0.80 vs. −1.86), Asia (−1.13 vs. −2.17), and North America (−0.80 vs. −1.72). This suggests that geographic information systematically moderates the models' negative evaluation of urban liveliness. The Boring attribute also shows a similar aggregate-level tendency, but in the opposite direction. Geo-referenced scores are lower than non-geo-referenced scores across all continents, including Oceania, Asia, South America, and North America. Since lower scores for Boring indicate a more favorable perception, this pattern suggests that geo-information often makes models evaluate places as less boring, particularly in regions such as Oceania and Asia.

In contrast, attributes such as Safe and Wealthy show more region-specific patterns. For Safe, geo-referenced scores are higher in Asia (3.14 vs. 2.67), Europe (2.92 vs. 2.73), and Oceania (approximately 4.95 vs. 3.88), but lower in Africa (1.25 vs. 1.98), North America (2.70 vs. 2.92), and South America (2.42 vs. 3.23). This indicates that safety perception is highly

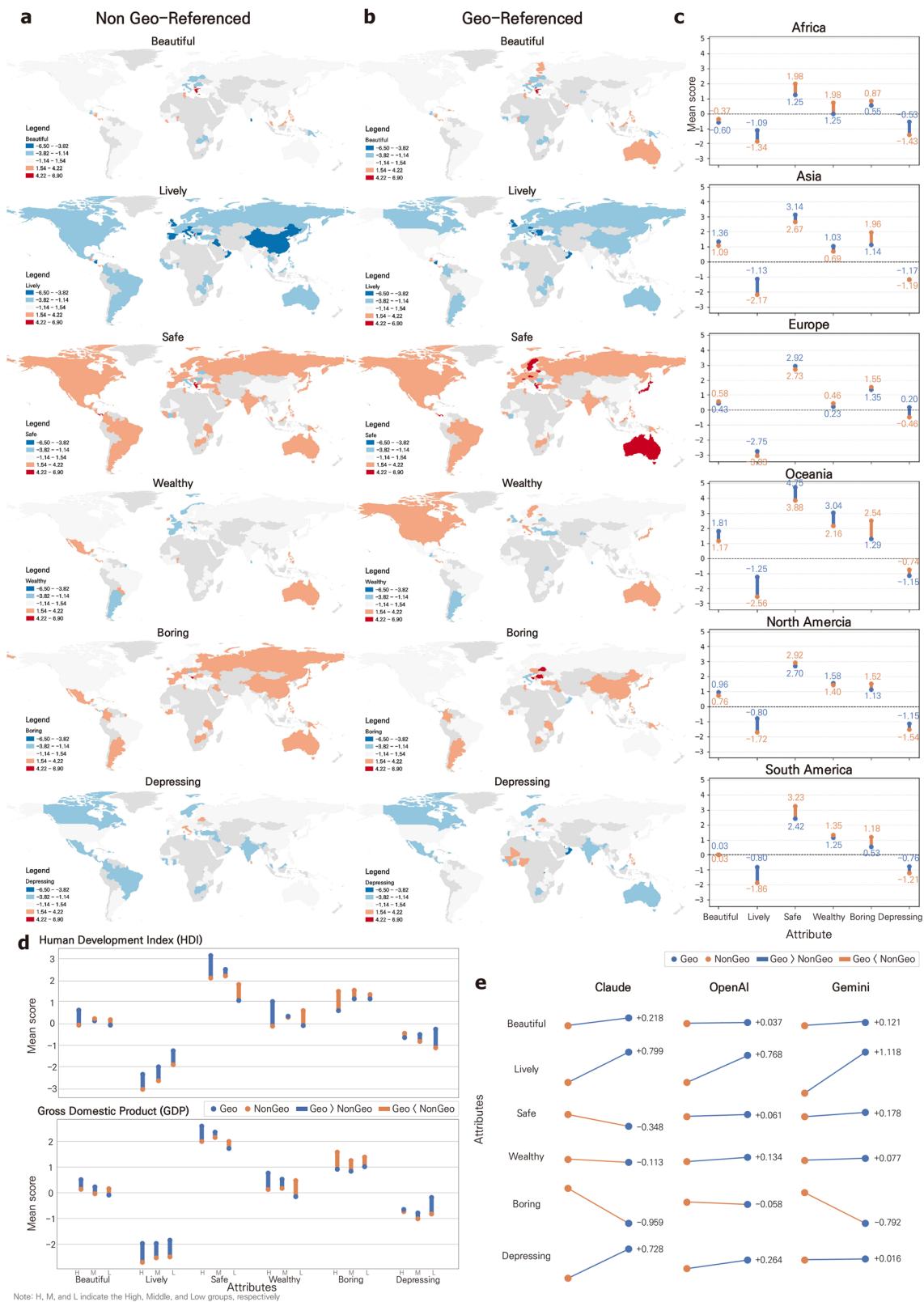


Fig. 3 | Perceptual bias in MLLMs with and without geographic information. **a** Country-level perceptual scores are shown for the non-geo-referenced condition. **b** Country-level perceptual scores are shown for the geo-referenced condition. **c** Continent-level mean scores are compared between geo-referenced and non-geo-referenced conditions across perceptual attributes. **d** Mean scores are compared across Human Development Index and Gross Domestic Product groups. H, M, and

L indicate High, Middle, and Low groups, respectively, based on tertile classification within the analytical sample. **e** Model-wise comparisons show attribute-level differences between geo-referenced and non-geo-referenced conditions for Claude, OpenAI, and Gemini. MLLM denotes multimodal large language model; HDI denotes Human Development Index; GDP denotes Gross Domestic Product.

sensitive to geographic context, but the direction of bias depends on the region. Similarly, wealth increases with geo-referencing in Asia, North America, and Oceania, while it decreases or remains lower in Africa, Europe, and South America.

Overall, these results suggest that geo-information does not simply raise or lower all perceptual scores in the same direction. Instead, it selectively affects certain perception dimensions. Lively and Boring show the most consistent cross-continental changes, whereas Safe and Wealthy reveal stronger regional differentiation. This indicates that geographic bias in MLLMs is both attribute-specific and region-specific, reflecting place-based priors rather than random variation. Therefore, the observed shifts are meaningful not only because of their magnitude, but also because they follow systematic patterns across perception dimensions.

To examine whether these continent-level perceptual patterns were consistently observed across cities, we further analyzed the distribution of city-level score differences for each attribute and model (Figure S3 and Table S3 in the Supplementary Information). The results reveal meaningful within-continent heterogeneity, particularly for some model–attribute combinations such as Boring and Depressing. However, several key directional patterns remain visible across many cities, especially for Lively and Safe. Accordingly, the continent-level estimates should be interpreted as central tendencies rather than uniformly shared effects across all cities.

At the aggregate continental level, the perceptual gaps between geo-referenced and non-geo-referenced prompts were smallest in Europe and largest in Africa. This suggests that in Europe, there is minimal divergence between the model's interpretation of SVI with and without geo-information, indicating that SVI effectively captures contextual information and that the MLLMs exhibit relatively low geographic bias in this region. However, because this divergence is measured against global perceptual labels, which are themselves biased against local perceptions⁴², both the perceptual model and the MLLMs may share a similar skewness, particularly in Africa. The larger discrepancies observed in Africa further raise questions about their deviation from local ground-truth perceptions, warranting future investigation.

We next compared the mean scores across groups categorized by HDI and GDP, following the same approach used in the instance detection task (Fig. 3d). The analysis revealed clear differences across High (H), Middle (M), and Low (L) groups for both HDI and GDP. In high-development or high-income countries (H), a strong positive bias emerged, with positive attributes such as Beautiful, Lively, Safe, and Wealthy were rated higher with geo-referencing, while negative attributes such as Boring and Depressing were rated lower. Conversely, in low-development or low-income countries (L), a distinct negative bias was observed, with positive attributes rated lower and negative attributes rated higher under geo-referencing conditions. Specifically, in countries with high HDI, the inclusion of geographic information increased evaluations of Beautiful, Lively, Safe, and Wealthy while decreasing Boring and Depressing. In contrast, in low-HDI countries, Beautiful, Safe, and Wealthy were rated lower and Depressing higher with geo-referencing. A similar pattern was observed for GDP, with high-GDP countries showing stronger positive perceptions and low-GDP countries displaying more negative perceptions when geo-information was provided. These findings suggest that development level influences not only the amplification of positive perceptions but also the differential reinforcement of negative attributes, and that geo-referencing accentuates these biases.

Additionally, the TOST procedure indicates that most perceptual attributes across the High, Middle, and Low groups satisfy the equivalence condition (Table S4 in the Supplementary Information). At the group level, the Middle and Low groups exhibit particularly consistent equivalence across attributes, while the High group shows a few mixed cases, though the overall differences remain small. These findings indicate that perception scores are largely stable regardless of whether countries are grouped by HDI or GDP. Overall, the results provide strong statistical evidence that differences between HDI- and GDP-based classifications are negligible, supporting our interpretation that the choice of grouping scheme does not substantively affect the observed perception patterns.

When comparing across models, Claude and Gemini exhibited stronger geographic bias, often producing overly positive responses for attributes such as Safe and Boring (Fig. 3e). These models tended to inflate such attributes when provided with geo information. In contrast, OpenAI's GPT-4o generally produced more balanced aggregate-level outputs across attributes, with less extreme directional shifts. Nevertheless, all three MLLMs systematically overvalued Lively while undervaluing Boring after providing geo information, revealing a pervasive underlying bias in MLLMs that transcends platform-specific architectures.

Collectively, these results support our hypothesis that geographic information systematically shapes MLLM perceptions, amplifying existing biases and reinforcing global stereotypes—thereby affirming the necessity of geographic red-teaming to audit and mitigate these effects in multimodal AI systems.

Exploration of perceptual bias

To further investigate the perceptual bias identified in the previous section, we conducted two additional tests: (1) a spatial scale sensitivity analysis to examine how different types of geo-referenced information influence model outputs, and (2) an embedding-based analysis to assess whether such biases are encoded in the models' latent semantic representations.

First, to examine which types of geo-referenced information contribute to the observed effects, we performed a sensitivity analysis across different spatial scales. The combined condition corresponds to the prompt used in the Perceptual attributes section, where coordinates, city, country, and continent information are provided simultaneously, while the remaining conditions present each type of geographic information separately. The results are presented in Fig. 4.

As shown in Fig. 4a, the distributions of perceptual differences are largely consistent across spatial scales, with all geo-referenced conditions exhibiting a positive shift relative to the baseline. However, the magnitude of this shift varies slightly. The largest difference in central tendency was observed between the combined and country conditions, reaching 0.108 for the mean and 0.229 for the median, equivalent to 17.9% and 27.3% of their maximum values, respectively. This indicates that single-scale conditions produce broadly comparable shifts, while the combined condition shows a slightly stronger upward shift. Notably, the combined condition—where multiple geographic cues are provided together—shows a relatively stronger upward shift compared to individual-scale inputs, suggesting a potential amplification effect. In contrast, single-scale conditions (coordinates, city, country, continent) display more comparable distributions, with only minor differences in central tendency. Overall, the spread and interquartile ranges remain similar across conditions.

Figure 4b provides a more detailed attribute-level comparison. The results reveal strong and consistent directional patterns across all spatial scales. While the overall direction of change remains consistent across conditions, subtle differences emerge depending on whether the information is combined or provided at a single scale. In particular, attributes such as Beautiful, Lively, Wealthy, and Boring have shown the combined condition tends to exhibit a more pronounced positive bias compared to single-scale inputs.

Importantly, the magnitude of these effects remains remarkably stable across different spatial scales. Differences between coordinates, city, country, and continent conditions are minimal, with no clear monotonic trend observed. This suggests that the perceptual shifts are not driven by a specific type of geographic input, but rather by the presence of geo-referenced information itself. However, a different pattern emerges when multiple forms of geographic information are introduced simultaneously. The combined condition—where coordinates, city, country, and continent are all provided—tends to produce more pronounced shifts in perception.

Taken together, these findings suggest that the key mechanism is not the specific scale of geographic information, but rather the activation of geo-referenced context itself, which can introduce systematic perceptual bias. Moreover, the results imply that providing multiple layers of geographic information may reinforce or amplify such bias, as different spatial cues

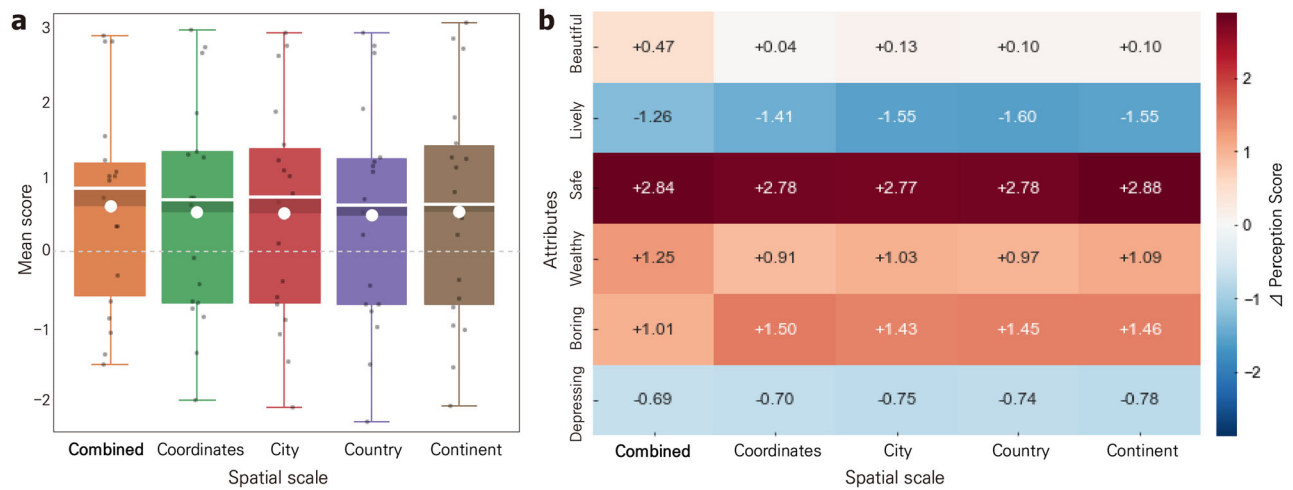


Fig. 4 | Sensitivity of perceptual bias to geospatial scale. **a** The distribution of mean perceptual differences is compared across five spatial-scale conditions: combined, coordinates, city, country, and continent. The white line indicates the median, and the white point indicates the mean. **b** Attribute-level mean differences are shown

across spatial-scale conditions. Warmer colors indicate higher perception scores, while cooler colors indicate lower perception scores. The combined condition includes coordinates, city, country, and continent information together.

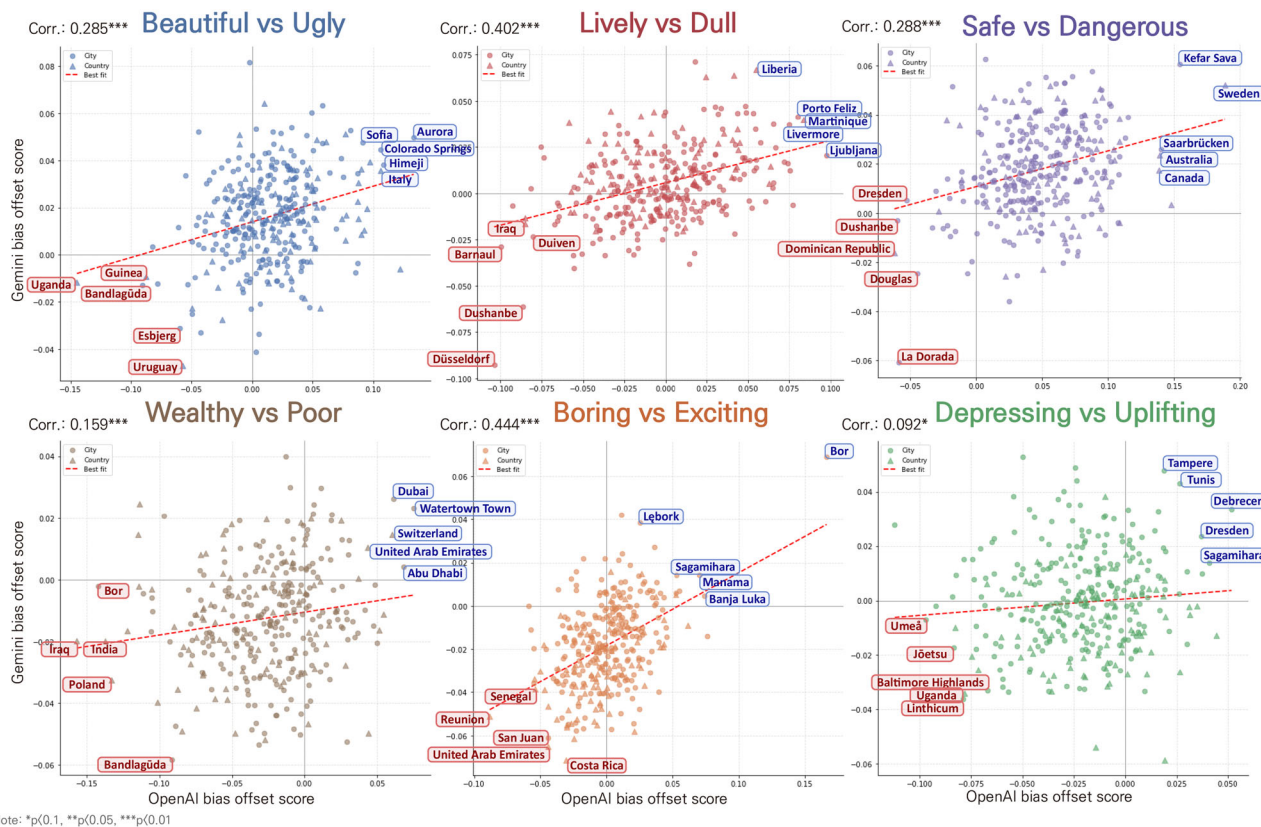


Fig. 5 | Embedding-based perceptual bias in OpenAI and Gemini. Scatter plots compare OpenAI and Gemini bias offset scores across six perceptual dimensions. Each point represents a city or country, and red dashed lines show linear trends.

Positive scores indicate stronger association with the positive perceptual pole, while negative scores indicate an association with the opposite pole.

collectively trigger place-based associations within the model. In this sense, perceptual bias appears to be linked to the presence and richness of geographic context, rather than to any single spatial scale alone.

Second, following recent taxonomies in bias evaluation¹⁶, we extended our analysis to the embedding level (Fig. 5). As a complement to the generated text-based analysis, we conducted an embedding-based experiment

to capture biases encoded in the models' latent representations, thereby assessing whether the observed perceptual biases are rooted not only in generation processes but also in the underlying semantic structure of the models.

Figure 5 represents an embedding-based bias across the two models. The x-axis represents the OpenAI Bias Offset Score, and the y-axis

represents the Gemini Bias Offset Score. Each point represents a geographic location's position on a semantic scale. A positive score means the model associates that place with the positive pole on the perception dimensions (Beautiful, Boring, Depressing, Lively, Safe, Wealthy), while a negative score indicates an association with the opposite pole (Ugly, Exciting, Uplifting, Dull, Dangerous, Poor). Each plot displays the bias on one dimension. The top five and bottom five place names are labeled in Fig. 5. For example, "Italy" and "Aurora" are close to "Beautiful"; "Dubai" and "Abu Dhabi" are very semantically close to "Wealthy". These findings confirm that the biases observed in our perceptual tasks are not only caused by the generation process or the prompting strategy but are deeply encoded in the models' latent spaces. Furthermore, six dimensions all demonstrate the positive correlation of the bias in Gemini and the OpenAI embedding model. Particularly, Boring vs. Exciting ($R^2 = 0.444$) and Lively vs. Dull ($R^2 = 0.402$) show the strongest correlations, which means the embedding-based bias on these two dimensions is highly consistent across these two models.

Discussion

This study explored how MLLMs interpret and respond to SVI from diverse geographic contexts. Through a dual-track evaluation of objective instance detection and subjective perceptual attributes, we systematically assessed performance variations across cities, continents and models. Our findings offer empirical insights into the presence and nature of geographic bias in MLLMs.

First, MLLMs' instance detection performance was influenced more by the model itself than by geographic location. Given that object detection is a largely objective task with limited subjective input, our results revealed negligible geographic variation in detection performance, indicating that bias-related concerns commonly raised in perceptual or semantic tasks may not apply here. The accuracy of instance detection was relatively stable across cities and continents, and performance was more strongly influenced by model architecture than by regional identity. For example, African cities, which are often underrepresented in training datasets, demonstrated some of the highest agreement scores. We suggest that these results may be attributed to the lower visual complexity of streetscapes in these regions. As a result, these findings suggest that MLLMs are increasingly competent at handling objective urban analysis tasks, regardless of location.

Second, the inclusion of geographic information significantly altered model visual perceptions, revealing geographic biases in subjective evaluations. While instance detection was largely consistent, we observed pronounced aggregate-level variation in perceptual attribute evaluations when geographic information was included in prompts. Notably, images associated with Africa and South America were rated less positively in dimensions like Wealthy and Beautiful after geographic context was introduced. In contrast, scenes from Asia, Oceania, and North America tended to see more positive perceptual scores. Extending this analysis, we compared mean scores across groups categorized by HDI and GDP, which were used as complementary indicators of developmental and economic context. Interestingly, there was a clear stratification. In high-development or high-income contexts, geo-referencing amplified positive bias, with positive attributes such as Beautiful, Lively, Safe, and Wealthy rated higher and negative attributes such as Boring, Depressing rated lower. Conversely, in low-development or low-income contexts, the opposite trend was observed, with diminished ratings for positive attributes and heightened ratings for negative ones under geo-referencing. These findings provide empirical evidence supporting prior concerns about the presence of geographic bias within LLMs^{10,35,43} and are also aligned with over- and underestimation of positive and negative indicators, respectively, by general-purpose perception models⁴². Specifically, the observed discrepancy confirms that geographic information acts as a semantic modifier, shaping the model's evaluation in ways that reflect implicit biases. These outcomes highlight the susceptibility of MLLMs to reproduce spatial stereotypes⁴⁴, particularly in perception tasks.

Lastly, the proposed framework extended current red-teaming methods on human-centric bias, like age, race, and gender¹⁶ to geography and

urban context. By combining multimodal evaluation, controlled prompt design, and balanced sampling across continents, the framework successfully exposed subtle yet systemic patterns of perceptual bias for diagnosing geographic bias in MLLMs. Importantly, we stress that the proposed framework is robust in the sense that it remains applicable across different datasets and geographic contexts. Specifically, robustness is demonstrated through the framework's consistent performance across six continents with heterogeneous urban environments, its stability under controlled prompt variations, and its balanced sampling strategy that mitigates continental representation bias. The framework encompasses data preprocessing, model analysis, evaluation methods, and interpretation, and is designed to remain applicable even as radically new models emerge in the future. In this sense, its contribution extends beyond the specific models tested here: it provides a methodological foundation that will remain relevant for subsequent generations of MLLMs.

From the perspective of model developers, our study offers actionable strategies for mitigating geographic bias. Hovy & Prabhmoey²⁵ identify five key sources of bias in NLP: data, annotation process, input representations, models, and research design. Our findings suggest that the observed disparities may be partly attributable to imbalanced or distorted training datasets^{45,46}, although this remains an assumption rather than a demonstrated causal factor, as the study did not explicitly test localized fine-tuning or dataset variation. As a countermeasure, we recommend that computer scientists and industry practitioners constructing MLLMs prioritize the development of more geographically representative training datasets and adopt continuous bias-monitoring protocols throughout the model lifecycle.

From the user's side, particularly for urban researchers and planners, our study demonstrates that both model selection and prompt design are critical factors in ensuring geographic fairness. Two critical insights emerged from our analysis: Model selection is consequential. Among the three tested platforms, OpenAI's GPT-4o exhibited the lowest level of perceptual bias at the aggregate level, producing more balanced evaluations across all continents and regions. In contrast, Claude and Gemini displayed a stronger positive bias toward higher HDI, GDP regions, while being more negative or inconsistent in their assessments of the lower regions. While we do not advocate its use unconditionally, these results suggest that GPT-4o may serve as a relatively reliable option for perceptual tasks, at least until newer models demonstrate superior performance. Therefore, model choice should be task-specific, considering not only cost and efficiency but also output reliability and measured geographic bias. This recommendation, with appropriate consideration of limitations and costs, could be seen as a practical contribution for both researchers and practitioners.

Second, prompt engineering demands caution. In tasks involving subjective evaluation, the inclusion of geographic information can unintentionally amplify bias. Our findings demonstrate that even when visual content is identical, the addition of place names or regional identifiers may skew model responses. Thus, geographic information should not be treated as a neutral prompt enhancement, nor should it be dismissed as inherently harmful. Rather, it should be treated as an analytical design choice whose value depends on the task. We recommend using non-geo-referenced prompts as a baseline and adding geographic information only when spatial context is analytically necessary. When included, geographic information should be provided at the minimum sufficient scale, as combined geo-referencing may amplify perceptual shifts. Therefore, prompt design in MLLM applications should be context-sensitive and bias-aware, particularly when deployed in settings that affect public perception, policy, or representation.

These insights underscore the importance of geographic red-teaming as a necessary component of responsible AI development. Without proactive interventions, such as geographic red-teaming, these systems risk reproducing existing socio-spatial inequities under the guise of algorithmic neutrality. As MLLMs continue to expand into domains such as urban analytics, public engagement, and automated decision-making, careful scrutiny of their spatial reasoning capabilities and biases will be essential to ensuring equity and fairness in AI deployment.

Based on the findings of this paper, we propose several important recommendations for future applications of MLLMs in urban planning research and practice: First, *be cautious when incorporating geographic identifiers into prompts for perceptual tasks*. Our results show that adding place names or regional markers can systematically alter model evaluations, sometimes providing contextual grounding but also potentially amplifying stereotypes in subjective perception tasks. Therefore, researchers and practitioners should first clarify whether the task requires visual-only assessment or context-aware interpretation. For visual-only comparisons, non-geo-referenced prompts can serve as a baseline; for context-aware tasks, geographic information may be included but should be explicitly justified, reported, and tested against a non-geo baseline. When included, geographic information should be provided at the minimum sufficient scale, as combined geo-referencing may amplify perceptual shifts. Where city- or region-level aggregation is used, variability should also be reported to distinguish general tendencies from patterns driven by a smaller subset of places.

Second, prioritize model selection and evaluation before deploying MLLMs in sensitive urban contexts. Different models exhibited divergent levels of perceptual bias, with GPT-4o producing more balanced outputs across continents, while others skewed toward high-HDI or high-GDP regions. Researchers should document the choice of models and consider comparative validation to ensure fairness. This practice is particularly important where model outputs may influence public perception, urban policy, or equity-related decision making. In addition, for sensitive perceptual tasks, future applications may benefit from multi-model validation or consensus-based approaches, where outputs are compared across models to distinguish robust patterns from model-specific bias. However, such ensemble strategies should be treated as a validation tool rather than an automatic solution, because shared agreement across models may also reflect common training-data biases.

Third, promote geographically representative training and continuous bias monitoring. Our evidence suggests that disparities are linked to imbalanced training datasets rather than inherent architectural flaws²⁵. Developers of MLLMs should expand datasets to better cover under-represented regions and implement systematic bias-checking protocols throughout the model lifecycle. This will help mitigate systemic inequities and foster fairer applications of MLLMs in global urban research and practice.

Overall, this study establishes a systematic Geographic Red-teaming framework to diagnose and quantify spatial biases in state-of-the-art Multimodal Large Language Models. By evaluating GPT-4o, Gemini 2.0 Flash, and Claude 3.7 Sonnet against a global dataset of over 200 cities, we identified a critical divergence in model performance between objective and subjective tasks. While instance detection, an objective task, remained remarkably stable and accurate across diverse geographic regions, urban perception assessments exhibited profound systemic biases. Specifically, the inclusion of geographic metadata (geo-referencing) can substantially act as a semantic modifier that significantly devalues images from Africa and South America while inflating scores for North America and Asia.

This study provides a global assessment of geographic bias in multimodal large language models (MLLMs) through urban perception and instance detection tasks using the Global StreetScapes dataset. We evaluated three leading MLLMs, Claude, Gemini, and OpenAI's GPT-4o, across objective and perceptual analyses. Results show that incorporating geo-referenced information can substantially reshape perceptual outputs, sometimes offering contextual grounding but also amplifying perceptual bias, while instance detection remained largely stable across regions. These variations reveal the models' tendency to reproduce spatial stereotypes, with aggregate-level negative bias toward African and South American cities and positive inflation for Asia and North America.

We, therefore, offer actionable strategies for mitigating geographic bias for both model providers and users. We urge model providers to promote geographically representative training and continuous bias monitoring; we remind users to avoid geographic identifiers in perceptual tasks.

Although this study offers a novel Geographic red-teaming framework, it is important to acknowledge limitations. This study is limited to specific model versions, and results may change with newer releases. The perceptual attributes are inherently subjective and may be interpreted differently across models. Following the Place Pulse framework, we did not impose strict definitions; however, such differences may partially contribute to the observed variation. In addition, the perceptual ground truth, derived from Place Pulse-based models, reflects aggregated global perceptions that may diverge from local viewpoints⁴². Future research should test localized datasets and broader model comparisons to further validate these findings. Furthermore, although our structured prompts and controlled settings were designed to mitigate hallucinated or unsupported outputs in MLLMs, hallucination remains an inherent limitation that should be considered when interpreting the results.

Methodology

Overview

This study proposes a multimodal evaluation framework to investigate geographic bias in MLLMs. We focus on two key analytical tasks in urban planning field: instance detection and perceptual attribute assessment, each targeting different dimensions of LLM performance.

In the instance detection task, MLLMs are evaluated on how accurately they can identify real-world built environment features from SVI. Bias is assessed in terms of how model results align with ground truth from a validated dataset -- Global StreetScapes⁴¹. The performance metric used is the agreement score, which quantifies the difference between detected instances and reference labels. In contrast, the perceptual attribute task evaluates subjective interpretations of the urban street environment. These attributes, Beautiful, Boring, Depressing, Lively, Safe, and Wealthy, are based on predicted perceptual scores from Global StreetScapes, which relies on Place Pulse 2.0^{47,48}, a widely adopted dataset that provides a validated, human-grounded baseline for assessing subjective urban perception. Importantly, this task includes a comparative experiment between geo-referenced and non-geo-referenced prompts, in order to assess whether the inclusion of geographic information such as country or city name systematically alters MLLM perceptions, thus serving as a mechanism for bias detection. By examining differences in model outputs across these conditions, this component seeks to uncover whether and how geographic context influences the models' perceptual judgments.

Three MLLMs are evaluated in this study: GPT-4o from OpenAI, Gemini 2.0 Flash from Google, and Claude 3.7 Sonnet from Anthropic. Each model was accessed by its API and tested under identical conditions for both tasks, enabling cross-model comparisons. Together, our research proposes an integrated framework for diagnosing potential geographic bias in MLLMs, addressing both factual detection and subjective interpretation within a globally representative urban dataset. The framework's robustness derives from its applicability across diverse geographic contexts and datasets sharing comparable visual and perceptual variables, encompassing data preprocessing, model analysis, evaluation methods, and interpretation, and is designed to remain applicable even as radically new models emerge in the future.

Data collection and processing

The primary data source for this study is the NUS Global StreetScapes dataset developed by Hou et al.⁴¹, which consists of approximately 10 million SVIs collected globally from crowdsourced platforms. Each image is annotated with a rich array of metadata, including geographic (e.g., city, country, coordinates), temporal (e.g., year, season, hour), contextual (e.g., driving surface, panoramic view), semantic (e.g., object labels), and perceptual attributes (e.g., subjective ratings based on human responses).

For consistency and analytical feasibility, we implemented a multi-stage preprocessing. First, we filtered the dataset to include only images from the years 2019 and 2020, which provided the largest subset of consistently available data. This temporal preprocessing was conducted to ensure a certain degree of streetscape consistency across the dataset. To control for

Table 2 | Evaluating prompt results

Criterion	Model					
	Claude		OpenAI		Gemini	
Structure / Format	X	Concise but a bit too simple	X	Minimal structure	✓	Highly structured
Role / Context Setting	✓	Defined through tone	X	Not defined	✓	Clear role definition
Examples Provided	✓	2 basic examples	X	None	✓	3 detailed multi-shot
Step-by-step Reasoning	X	Implied but not broken down	X	Not present	✓	Included explicitly
Output Format Specified	✓	Concise and correct	✓	Clearly included	✓	Clear, enforced format
Error Case Handling	✓	Clearly instructed	✓	Clearly instructed	✓	Clearly instructed

environmental variation, we further restricted the images to daytime hours (07:00–18:00), non-winter seasons, clear weather, and non-panoramic views. Furthermore, to increase consistency, only images sourced from the same platform (Mapillary) and captured from driving surfaces were retained. This preprocessing yielded a filtered dataset of 325,473 images covering 239 cities (Table S5 in the Supplementary Information). To address potential geographic imbalance and ensure cross-continental representativeness, we employed stratified random sampling, selecting 1000 images per continent, with the additional constraint that every city in the subset be represented by at least one image. The final curated dataset consists of 6000 images across 238 cities in 95 countries evenly distributed across six continents (Table S6 in the Supplementary Information). This balanced dataset enables fair model comparisons and robust insights into spatial equity across MLLM outputs. In addition, to examine differences in perceptual bias across spatial scales, we constructed a subset of 600 images from the original dataset (Table S7 in the Supplementary Information). Following the same sampling strategy, we randomly selected 100 images per continent using a fixed pandas random seed to ensure reproducibility, and the summary statistics of this subset are provided in the manuscript.

Prompt engineering

To ensure fairness and internal validity across model comparisons, we carefully conducted prompt engineering based on each model's official usage guidelines^{49–51}. The systematized criteria derived from these guidelines are presented in Table S8 in the Supplementary Information. Specifically, we adopted a two-stage prompting strategy to balance model-specific optimization and cross-model comparability. In the first stage, we conducted model-specific prompt engineering using each model's chat interface (GPT-4o, Gemini 2.0 Flash, and Claude 3.7 Sonnet), which differs from the API environment, following their recommended prompting practices. The resulting prompts were developed in accordance with each model's guidelines and are summarized in Table S9 in the Supplementary Information.

In the second stage, we systematically evaluated these candidate prompts (Table 2). Upon comparative assessment, the Claude 3.7-based prompt structure emerged as the most robust and structured. Importantly, this selected prompt was then uniformly applied to all three models via their APIs for both instance detection and perceptual tasks. Although each MLLM has unique architectural constraints, we designed a standardized prompt template and applied it uniformly across. This standardization ensures that any observed bias is not derived from variation in prompt design itself but reflect intrinsic model behavior.

The parameters of each prompt were configured based on the characteristics of each task. For the instance detection task, which involves factual recall of objects in an SVI, we set the temperature to 0 to eliminate variability and encourage deterministic outputs⁴⁹. Conversely, for the perceptual attribute task, which requires subjective interpretation of visual scenes, we used a temperature of 1, allowing the models to express a broader range of evaluative judgment. The max token parameter was set to 256, which is sufficient to generate complete responses for the given tasks. All models were accessed via their official APIs without any additional fine-tuning or modification, ensuring that each model was evaluated in its default

deployed state. To prevent carryover effects between responses, the model context was reset after each inference, ensuring that all predictions were generated independently.

For the instance detection task, prompts instructed the models to identify the presence or absence of ten predefined built environment instances in a given image. These object categories were selected based on both their high frequency in the Global Streetscapes dataset and their relevance to built environment auditing in prior urban studies⁸. Outputs were returned in binary format: 1 for present, 0 for absent, and 9 for invalid or indoor images. This format allowed for straightforward calculation of detection accuracy against ground truth labels. A portion of the prompt is shown below, with the full prompt available in Table S9 in Supplementary Information.

“You are an expert computer vision system specifically trained to identify common urban objects in images. Your task is to analyze the input image and determine the presence or absence of the following 10 objects:

1. Car, 2. Billboard, 3. Traffic sign, 4. Road, 5. Building, 6. Vegetation (trees, bushes, plants), 7. Sidewalk, 8. Streetlight, 9. Bicycle, 10. Trash can
<instructions>

Carefully examine the entire image for each object - Consider partial objects (even if only a portion is visible) - Be thorough and methodical in your analysis - First identify what's in the image overall, then check for each specific object
</instructions>

<output format>

For each object present in the image, respond with a 1; if an object is not present, respond with a 0. Ensure you separate the numbers with commas for data processing purposes. Be sure to sort out invalid images, such as empty photos and photos depicting indoors. In this case, respond 9 to denote the image error. Do not include any chat content in the response.
</output format>”

For the perceptual attributes task, two prompt variants were used. The first prompt excluded any geographic information, while the second explicitly included the city, country, and continent associated with the image. For both cases, the output was a set of six numerical ratings—Beautiful, Boring, Depressing, Lively, Safe, and Wealthy—scored on a 0–10 scale. A return value of 100 was designated to indicate cases where the image was invalid or an error occurred during processing. A portion of the two prompts is shown below, with the full prompts available in Table S10 in Supplementary Information.

“You are an expert urban perception analyst trained to evaluate human perceptions of urban environments in images. Your task is to analyze the input image and provide perception scores for six key dimensions:

<perception_dimensions>

1. Beautiful, 2. Boring, 3. Depressing, 4. Lively, 5. Safe, 6. Wealthy

</perception_dimensions>

<output_format>

Score each dimension from 0–10, with 0 being the lowest and 10 being the highest.

Present your final scores in a comma-separated format: Beautiful, Boring, Depressing, Lively, Safe, Wealthy

For example: 7,3,2,8,7,6

For invalid images (non-urban scenes, blank images, etc.), respond with "100" to denote the image error.

Do not include any chat content in the response.

</output_format>

"You are an expert urban perception analyst trained to evaluate human perceptions of urban environments in images. Your task is to analyze the input image from {city} in {country} in {continent} and provide perception scores for each dimension.

<location_information>

Coordinates: ({lat}, {lon})

City: {city}

Country: {country}

Continent: {continent}

</location_information>

For the additional analysis of perceptual bias across geospatial scales, we modified the geo-referenced prompt by adjusting only the component that provides geographic information. Specifically, separate prompt variants were constructed for each spatial scale (coordinates, city, country, and continent) while keeping all other elements of the prompt identical (Table S11 in the Supplementary Information).

Evaluation methods

Two distinct evaluation strategies were employed to assess the outputs of MLLMs. For the instance detection task, we adopted the agreement score metric, previously validated by Jang and Kim⁸. This metric quantifies the model's accuracy by comparing the presence or absence of each object in its output to the ground truth annotations provided by the Global StreetScapes dataset. The agreement score represents the proportion of correctly identified instances across the ten target instances and is computed per model and per image. This metric provides a straightforward and interpretable measure of objective detection performance.

For the perceptual attribute task, our evaluation focused on the mean score for each of the six perceptual dimensions. The mean score was derived using the same agreement score metrics employed in the instance detection task, calculated as the difference between the MLLM-generated rating and the corresponding rating from the Global StreetScapes dataset for each perceptual attribute and image. While the Global StreetScapes dataset does not represent ground truth, it serves as a reasonable baseline, as it was trained based on actual human survey data⁴⁷. Subsequently, we compare the mean scores generated from geo-referenced prompts with those from non-geo-referenced prompts to assess how the inclusion of geo-information influences model outputs. Finally, we interpret our results along three dimensions, which are continent, Human Development Index (HDI), and Gross Domestic Product (GDP). The HDI data were obtained from UNDP, while the GDP data were sourced from the World Bank Open Data.

HDI and GDP were used as complementary proxies for socioeconomic and developmental context, which may shape both the physical form of urban environments and the social meanings attached to places. HDI captures multidimensional development conditions, including health, education, and standard of living, whereas GDP represents the economic scale of each country. These indicators are also relevant to geographic bias evaluation because prior work has shown that LLM performance can vary systematically across world regions and income groups³⁴. Countries were classified into High, Middle, and Low groups using tertile-based cutoffs within our analytical sample for each indicator.

As a supplementary of generated text-based analysis, we also conducted an experiment to depict the embedding-based bias, which is encoded within the word representations of the models. We first fetched the embedding of each place (continent, country, or city) from OpenAI (text-embedding-3-large) and Gemini (gemini-embedding-001). Claude embedding model is not included because Anthropic does not offer an embedding model for public service. Each embedding is a vector with 3072 dimensions. The embedding bias is calculated according to following

formula Eq. (1)^{52,53},

$$\text{Bias Offset} = \cos(V_{\text{place}}, V_{\text{perceptual}}) - \cos(V_{\text{place}}, V_{\text{opposite}_{\text{perceptual}}}) \quad (1)$$

where $\cos$ indicates the cosine similarity between two vectors (embedding of the word). If the two elements are close in the embedding space, the cosine similarity will be close to 1, otherwise 0. By calculating the relative difference between two antonyms, we effectively cancel out this "background noise" (the general geographic weight of the city) to reveal the specific perceptual polarity. We then analyzed the correlation of bias between the two models across each dimension of perception.

In addition, to assess the stability of the perceptual outputs, we conducted a repeated-run analysis using a subset of 600 images. The perceptual assessment was repeated three times under the same prompt and parameter settings, and the run-to-run variability was compared across models and attributes (Table S12 in the Supplementary Information).

Data availability

Global StreetScapes data is from the paper(<https://doi.org/10.1016/j.isprsjprs.2024.06.023>) and provided within the [Github](#).

Received: 1 November 2025; Accepted: 10 August 2026;

Published online: 28 August 2026

References

- Zheng, Y. et al. Urban planning in the era of large language models. *Nat. Comput. Sci.* **5**, 727–736 (2025).
- Chen, A., Ge, X., Fu, Z., Xiao, Y. & Chen, J. TravelAgent: an AI assistant for personalized travel planning. Preprint at. arxiv.org/abs/2409.08069 (2024).
- Han, J., Zheng, Z., Lu, X. Z., Chen, K. Y. & Lin, J. R. Enhanced earthquake impact analysis based on social media texts via large language model. *Int. J. Disaster Risk Reduct.* **109**, 104574 (2024).
- Fu, X. et al. Large language models in urban planning. *Nat. Cities* **2**, 585–592 (2025).
- Luo, Y., Cao, Z., Jin, X., Liu, K. & Yin, L. Deciphering human mobility: inferring semantics of trajectories with large language models. in *Proc. 25th IEEE International Conference on Mobile Data Management*, 289–294 (IEEE, 2024).
- Zhang, Z., Amiri, H., Liu, Z., Zhao, L. & Züfle, A. Large language models for spatial trajectory patterns mining. in *Proc. 1st ACM SIGSPATIAL International Workshop on Geospatial Anomaly Detection* 52–55 (ACM, 2024).
- Li, Z., Xu, J., Wang, S., Wu, Y. & Li, H. StreetviewLLM: extracting geographic information using a chain-of-thought multimodal large language model. Preprint at. arxiv.org/abs/2411.14476 (2024).
- Jang, K. M. & Kim, J. Multimodal large language models as built environment auditing tools. *Prof. Geogr.* **77**, 84–90 (2025).
- Liang, X., Xie, J., Zhao, T., Stouffs, R. & Biljecki, F. OpenFACADES: an open framework for architectural caption and attribute data enrichment via street view imagery. Preprint at. arxiv.org/abs/2504.02866 (2025).
- Ki, D., Lee, H., Park, K., Ha, J. & Lee, S. Measuring nuanced walkability: leveraging ChatGPT's vision reasoning with multisource spatial data. *Comput. Environ. Urban Syst.* **121**, 102319 (2025).
- Malekzadeh, M., Willberg, E., Torkko, J. & Toivonen, T. Urban attractiveness according to ChatGPT: contrasting AI and human insights. *Comput. Environ. Urban Syst.* **117**, 102243 (2025).
- Lan, H. Interpretable multimodal framework for human-centered street assessment: integrating visual-language models for perceptual urban diagnostics. Preprint at. arxiv.org/abs/2506.05087 (2025).
- Tang, Y. et al. From street views to urban science: discovering road safety factors with multimodal large language models. Preprint at. arxiv.org/abs/2506.02242 (2025).

14. Crooks, A. & Chen, Q. Exploring the new frontier of information extraction through large language models in urban analytics. *Environ. Plan. B Urban Anal. City Sci.* **51**, 565–569 (2024).
15. Lindsey, J. et al. On the biology of a large language model. Transformer Circuits Thread. transformer-circuits.pub/2025/attribution-graphs/biology.html (2025).
16. Gallegos, I. O. et al. Self-debiasing large language models: zero-shot recognition and reduction of stereotypes. Preprint at arxiv.org/abs/2402.01981 (2024).
17. Blodgett, S. L. & O'Connor, B. Racial disparity in natural language processing: a case study of social media African-American English. Preprint at arxiv.org/abs/1707.00061 (2017).
18. Garrido-Muñoz, I., Montejó-Ráez, A., Martínez-Santiago, F. & Ureña-López, L. A. A survey on bias in deep NLP. *Appl. Sci.* **11**, 3184 (2021).
19. Gupta, V., Venkit, P. N., Wilson, S. & Passonneau, R. J. Sociodemographic bias in language models: a survey and forward path. Preprint at arxiv.org/abs/2306.08158 (2023).
20. Bolukbasi, T., Chang, K. W., Zou, J. Y., Saligrama, V. & Kalai, A. T. Man is to computer programmer as woman is to homemaker? Debiasing word embeddings. *Adv. Neural Inf. Process. Syst.* **29**, (2016).
21. Brown, T. et al. Language models are few-shot learners. *Adv. Neural Inf. Process. Syst.* **33**, 1877–1901 (2020).
22. Sap, M., Card, D., Gabriel, S., Choi, Y. & Smith, N. A. The risk of racial bias in hate speech detection. in *Proc. 57th Annual Meeting of the Association for Computational Linguistics* 1668–1678 (ACL, 2019).
23. Shearer, E., Martin, S., Petheram, A. & Stirling, R. Racial bias in natural language processing. Oxford Insights. oxfordinsights.com/wp-content/uploads/2024/07/SHARED_-Racial-Bias-in-Natural-Language-Processing.pdf (2019).
24. Floridi, L. & Chiriatti, M. GPT-3: its nature, scope, limits, and consequences. *Minds Mach.* **30**, 681–694 (2020).
25. Hovy, D. & Prabhumoye, S. Five sources of bias in natural language processing. *Lang. Linguist. Compass* **15**, e12432, <https://doi.org/10.1111/Inc3.12432> (2021).
26. Johannsen, A., Hovy, D. & Søgaard, A. Cross-lingual syntactic variation over age and gender. in *Proc. 19th Conference on Computational Natural Language Learning* 103–112 (ACL, 2015).
27. Abid, A., Farooqi, M. & Zou, J. Persistent anti-Muslim bias in large language models. Preprint at arxiv.org/abs/2101.05783 (2021).
28. Manzini, T., Lim, Y. C., Tsvetkov, Y. & Black, A. W. Black is to criminal as Caucasian is to police: detecting and removing multiclass bias in word embeddings. Preprint at arxiv.org/abs/1904.04047 (2019).
29. Mirza, S., Coelho, B., Cui, Y., Pöpper, C. & McCoy, D. Global-liar: factuality of LLMs over time and geographic regions. Preprint at arxiv.org/abs/2401.17839 (2024).
30. Faisal, F. & Anastasopoulos, A. Geographic and geopolitical biases of language models. Preprint at arxiv.org/abs/2212.10408 (2022).
31. Lu, J. G., Song, L. L. & Zhang, L. D. Cultural tendencies in generative AI. *Nat. Hum. Behav.* **9**, 2360–2369 (2025).
32. Sukiennik, N., Gao, C., Xu, F. & Li, Y. An evaluation of cultural value alignment in LLM. Preprint at arxiv.org/abs/2504.08863 (2025).
33. Manvi, R., Khanna, S., Burke, M., Lobell, D. & Ermon, S. Large language models are geographically biased. Preprint at arxiv.org/abs/2402.02680 (2024).
34. Moayeri, M., Tabassi, E. & Feizi, S. Worldbench: quantifying geographic disparities in LLM factual recall. in *Proc. 2024 ACM Conference on Fairness, Accountability, and Transparency*, 1211–1228 (ACM, 2024).
35. Kim, J., Lee, J., Jang, K. M. & Lourentzou, I. Exploring the limitations in how ChatGPT introduces environmental justice issues in the United States: a case study of 3,108 counties. *Telemat. Inform.* **86**, 102085 (2024).
36. Renshaw, A., Lourentzou, I., Lee, J., Crawford, T. & Kim, J. Comparing the spatial querying capacity of large language models: OpenAI's ChatGPT and Google's Gemini Pro. *Prof. Geogr.* **77**, 186–198 (2025).
37. Ge, S. et al. MART: improving LLM safety with multi-round automatic red-teaming. Preprint at arxiv.org/abs/2311.07689 (2023).
38. Liu, Y. et al. Trustworthy LLMs: a survey and guideline for evaluating large language models' alignment. Preprint at arxiv.org/abs/2308.05374 (2023).
39. Sarkar, U. E. Evaluating alignment in large language models: a review of methodologies. *AI Ethics* **5**, 3233–3240 (2025).
40. Hong, Q., Zhao, H., Jiang, Y., Cheng, C. & Long, Y. Measuring neighbourhood-level perceived residential environment quality (PREQ) from multi-platform user-generated text: an LLM-enhanced framework. *Habitat Int.* **176**, 103944 (2026).
41. Hou, Y. et al. Global Streetscapes—a comprehensive dataset of 10 million street-level images across 688 cities for urban science and analytics. *ISPRS J. Photogramm. Remote Sens.* **215**, 216–238 (2024).
42. Quintana, M. et al. Global urban visual perception varies across demographics and personalities. *Nat. Cities* **2**, 1092–1106. <https://doi.org/10.1038/s44284-025-00330-x> (2025).
43. Tao, Y., Viberg, O., Baker, R. S. & Kizilcec, R. F. Cultural bias and cultural alignment of large language models. *PNAS Nexus* **3**, 346 (2024).
44. Tan, Y. C. & Celis, L. E. Assessing social and intersectional biases in contextualized word representations. *Adv. Neural Inf. Process. Syst.* **32**, 13230–13241 (2019).
45. Kim, H. & Lee, S. Analysis of changes in urban space usage for alleviating depression before and after COVID-19: application of KoBERTopic model with social media big data. *J. Urban Des. Inst. Korea* **25**, 109–125 (2024).
46. Jang, K. M. et al. Place identity: a generative AI's perspective. *Humanit. Soc. Sci. Commun.* **11**, 1–16 (2024).
47. Dubey, A., Naik, N., Parikh, D., Raskar, R. & Hidalgo, C. A. Deep learning the city: quantifying urban perception at a global scale. in *Proc. European Conference on Computer Vision* 196–212 (Springer International Publishing, 2016).
48. Salesses, P., Schechtner, K. & Hidalgo, C. A. The collaborative image of the city: mapping the inequality of urban perception. *PLoS ONE* **8**, e68400 (2013).
49. OpenAI. Best practices for prompt engineering with the OpenAI API. OpenAI Help Center. help.openai.com/en/articles/6654000-best-practices-for-prompt-engineering-with-the-openai-api (2025).
50. Google A. I. for Developers. Prompt design strategies. Google AI for Developers. ai.google.dev/gemini-api/docs/prompting-strategies (2025).
51. Anthropic. Prompt engineering overview. Anthropic Documentation. docs.anthropic.com/en/docs/build-with-claude/prompt-engineering/overview (2025).
52. Caliskan, A., Bryson, J. J. & Narayanan, A. Semantics derived automatically from language corpora contain human-like biases. *Science* **356**, 183–186 (2017).
53. Guilbeault, D., Delecourt, S. & Desikan, B. S. Age and gender distortion in online media and large language models. *Nature* **646**, 1129–1137 (2025).

Acknowledgements

This paper was presented at AI and Cities Conference, under the title “Exploring Physical and Perceptual Bias in Multimodal LLMs: Evidence from the Global Streetscapes”. We would like to express our sincere gratitude to all those who provided constructive feedback.

Author contributions

H.K. and L.X. formulated the initial concept, L.S. and F.B. revised the initial concept. M.Q. and F.B. curated a data resource. H.K., L.X., and M.Q. reviewed the literature and wrote the draft. H.K. created the figures, edited and reviewed the manuscript. M.Q., F.B., and L.S. contributed to the methodology, edited, and reviewed the manuscript. L.S. and M.Q. supervised the manuscript. All authors have read and approved the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at

<https://doi.org/10.1038/s42949-026-00466-2>.

Correspondence and requests for materials should be addressed to Sugie Lee.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.